

Project Proposal: Specifying Real-World Data Use For Trial Acceleration

András Novoszáth, 2026 <https://www.linkedin.com/in/andrasnovoszath>

Proposed Project

Specifying RWD use for trial acceleration.

Trial duration & costs are blockers in aging R&D. Strengthening RWD could decrease trial duration, cost, and evidence uncertainty. Downstream, it could speed up iteration and lower barriers to entry.

Outcome - **An RWD usability framework:**

- How RWD can accelerate longevity trials
- Criteria for whether a dataset or workflow is fit for regulatory-grade use

Career Goal

- **Evidence acceleration with RWD:** My goal is to accelerate evidence generation in longevity R&D by strengthening the use of real-world data.
- **Trial-burden reduction:** My operational aim is to reduce the time, cost, and uncertainty of Phase II evidence generation by enabling credible external controls and better-informed trial design.
- **Practical translation:** Over time, I want this to translate into tools, programs, and possibly company creation. The immediate objective is to create a reusable framework for evidence generation.

Project Fit

Why This Problem Matters

- **Longevity relevance:** Longevity trials face long time horizons, difficult endpoints, heterogeneous populations, and high evidentiary uncertainty. Better use of RWD reduces the dependence on control-arm evidence, decreases trial duration and cost, and improves evidence usability.
- **RWD gap:** The limiting factor is not always the lack of data, but the lack of fitness, reusability, and trustworthiness of the data available.
- **Core pipeline leverage:** This topic targets evidence generation itself, a bottleneck of the full evidence pipeline. It can have cumulative effects in trial design, regulatory interaction, capital allocation, portfolio decisions, and overall ecosystem growth.
- **Leverage:** The output would be a reusable resource that helps others specify and operationalize better evidence pathways.

Why Now

- **Regulatory movement:** FDA and EMA have further opened towards the use of RWD/RWE, with emphasis on data quality, traceability, and fitness for purpose.
- **Standardization opportunity:** Regulatory-grade use increasingly depends on data standards and specifications rather than volume alone.

- **Longevity timing:** In longevity, where trials are slow and endpoints are difficult, early improvements in evidence strategy can produce outsized gains.

Value Produced

By Primary Stakeholders

- **Clinical and biostatistics leadership at longevity startups:** Guidance on where RWD can credibly reduce trial burden or improve design choices.
- **Researchers and trial designers:** Better criteria for choosing datasets, evidence pathways, and methodological approaches.
- **Founders and executives:** Better evidence strategy for deciding which trial paths are worth pursuing and where uncertainty can be reduced.

Across The Longevity R&D Pipeline

- **Program selection:** Better evidence standards reduce uncertainty around programs.
- **Trial design:** Better RWD improves cohort construction, external controls, eligibility strategy, endpoint selection, and feasibility assessment.
- **Regulatory readiness:** Better traceability and fitness criteria improve the credibility of evidence packages.
- **Iteration speed:** RWD use can shorten trial duration and increase feedback loops.
- **Access:** Lower evidence-generation burden makes studies more feasible for smaller or earlier-stage projects.

Competence Fit

Why I Fit This Problem

Evidence Systems, Data Engineering

This project needs to consider how data becomes usable evidence.

- **Production delivery:** I have successfully delivered data science, machine learning, analytics, and data engineering work across enterprise, research, startup, and frontier-data settings. The core contribution was always to create 'value' (e.g., insights, analysis, models, tools) from data.
- **Methodological training:** My PhD research in science and technology studies provided an in-depth description of how knowledge is produced, stabilized, and made credible through methods, infrastructures, and institutions.

Clinical Evidence And Methodology

This project depends on familiarity with clinical research, and especially the type of data used there.

- **Clinical trial reporting:** At Bold Type Consulting, I built a reporting pipeline from clinical trial data assessing a medical device and generated insights that informed investment decisions.
- **Health time-series modeling:** At Bioepic, I developed a glucose forecasting model whose predictive performance matched market-leading commercial medical devices.
- **Clinical transition:** I am extending this base toward clinical data models and RWD methodology (e.g., OMOP/OHDSI).

Regulation, Traceability, And Implementation

Regulatory-grade evidence depends on data quality, traceability, and fitness for purpose. The project requires connecting formal requirements to usable systems.

- **Governance context:** In Diligent, I have worked in governance, risk, and compliance environments with strong requirements around auditability, security, and data governance.
- **MiFID II experience:** I turned MiFID II security compliance requirements into software specifications in a private banking setting.
- **Traceability mindset:** I have worked with systems where provenance and traceability shaped whether outputs could be trusted and used.

Economics, Incentives, And Business Translation

A durable RWD solution should take into account incentives, market structure, and how value is delivered.

- **Economic grounding:** My economics background supports analysis of incentives, coordination, market structure, and resource allocation.
- **Knowledge-effect in markets:** In my research, I highlighted how technologies (i.e., a special-purpose currency) shape markets via their knowledge effects.

Turning Ambiguous, High-Friction Systems Into Operationalizable Frameworks

This project is valuable if it can turn fragmented information, mixed standards, and abstract constraints into an operationalizable framework.

- **Delivering on messy data:** Throughout my academic and professional work, I have repeatedly delivered solutions based on data that was incomplete, unstable, weakly documented, or poorly suited for downstream use.
- **Cross-disciplinary delivery:** I repeatedly engaged with different disciplines (e.g., economics, social sciences, data science, business, software engineering) and modalities (e.g., qualitative fieldwork, high-level analysis, data science, operational framing, and software development) and drew upon their synergies.
- **Structured communication:** As a technical content writer, I have delivered content about specialized data & AI topics (e.g., database management, machine learning operations, blockchain data) many times.

Expected Outputs

Minimal Delivery

- **RWD usability framework for longevity trials:** A framework for where and how RWD can accelerate longevity trials. It will discuss criteria for external controls, cohort construction, causal approaches, and study feasibility. It will provide guidance about whether a dataset or workflow is fit for regulatory-grade use.

Possible Extensions

- **Trial-efficiency synthesis:** A focused synthesis of trial methodology and process-improvement work tailored to longevity.

- **Structured resource map:** A curated map of relevant datasets, tools, trial designs, and methodological gaps.
- **Incentive layer:** A clearer account of the economic and organizational constraints that shape adoption.
- **Machine-readable format:** A structured version that supports comparison, updating, and tooling.
- **AI-ready layer:** Further criteria for identifying which RWD assets are not only regulatory-relevant but also suitable for downstream AI use.